

Data X , groups S_g

Random walk P

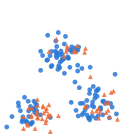
P-RW operator $P_{(\nu)}$

Entropy $\hat{H}(t)$

Stage 1: $Z \leftarrow \Pi_{\pi}(P_{(\nu)}Z)$

k -means on Z

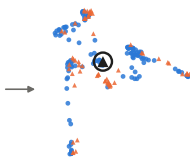
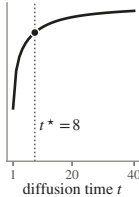
Stage 2: fair assignment



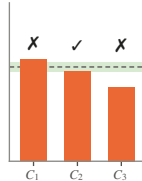
not reversible



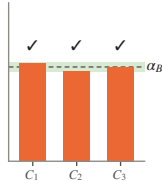
$D_{\pi}P_{(\nu)}$ symmetric



$F^{\top}Z=0$: centroids coincide



balance 0.263



balance 0.319

● group A ▲ group B

operators: block averages over (cluster, group); darker = larger

bars: share of group B per cluster; shaded: range allowed by (P_{δ}) ; ✓ / ✗ in / out